Catherine Lin

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EDUCATION

Bridgeland High School (CEEB: 440281) Cypress, Texas

Cumulative GPA: 6.8298/7
Non-Weighted GPA: 3.9556/4
ACT: 36 Composite (English: 35, Math: 36, Reading: 36)
Rank: 29/887

RESEARCH

Science Internship Program (SIP) under University of Santa Cruz Santa Cruz, CA

CPM-13: AI assistants for digital boundary communication and management Jun 2024 – Aug 2025

- Conducted Human-Computer Interaction (HCI) research on AI-powered digital boundary management systems, reviewing literature and analyzing strategies to improve user well-being through intelligent notification filtering and prioritization;
- Collaborated with a research team to design and prototype AI assistant concepts that help users establish, manage, and communicate digital boundaries, promoting healthier technology use and work-life balance;
- Continued the project beyond the internship as a co-author on a research paper, "The Effectiveness of AI Mediator Chatbots," currently in preparation for submission to the CHI 2027 Conference

Pioneer Academics Jun 2025 – Aug 2025

- Elected to participate in Pioneer Academics, a selective international undergraduate-level research program, under the mentorship of a university professor.
- Conducted independent research on how artificial intelligence is reshaping workforce readiness for computer scientists across different career stages, analyzing trends from the pre-AI era through future workforce projections.
- Synthesized findings from academic literature, industry reports, and workforce datasets to evaluate changes in skill requirements, job market accessibility, AI adoption, and the need for continuous upskilling.

Research under Professor Suleyman Uludag, University of Michigan Jun 2025 – Present

The Impact of Artificial Intelligence on the Computer Science Job Market

- Synthesized evidence from 40 academic publications, industry reports, labor economics studies, and human–AI interaction research to identify workforce trends across entry-, mid-, and senior-level careers.
- Analyzed how AI is shifting professional responsibilities from routine task execution toward AI integration, evaluation, governance, ethical oversight, and human-centered decision-making.
- Authored a structured narrative literature review with Professor Süleyman Uludağ that is currently being prepared for submission to a conference.

INDEPENDENT PROJECT

***Cofounder & CEO*, Sound of the Future** May 2025 – Present

- Founded a nonprofit promoting hearing protection and advocating for the Deaf and hard of hearing community by creating educational content on hearing health, safe listening practices, and hearing-loss prevention through social media and podcasts; expanded the organization to 5 chapters across the US; expanded the organization to 2 national chapters across the globe.
- Lead educational campaigns, podcasts, school/community programming, and digital outreach focused on safe listening practices and hearing accessibility.
- Lead fundraising and community initiatives supporting hearing-advocacy organizations including **HLAA**, while representing the organization in national and international innovation opportunities.
- Build partnerships with schools, nonprofits, healthcare professionals, and community organizations; represented the organization on **KPFT Houston** to discuss hearing protection and youth advocacy.
- Secure funding opportunities through scholarship and grant applications and represent Sound of the Future at major events, including **STEMGO Pro Fest and TechFest**, to educate attendees, recruit volunteers, and expand the organization's reach.

***Independent Research*, AI & Employment Research Initiative** May 2026 – Present

- Conduct structured interviews with HR professionals and industry employees examining AI adoption, hiring decisions, job displacement, and emerging workforce skills.
- Designed interview instruments evaluating AI adoption, hiring decisions, job displacement, and emerging workforce skills.
- Analyze qualitative interview data to identify how AI is changing employment expectations and workforce preparation: presented on a self coded public website

***Independent Research,* AI & Asian Restaurant Technology Study** June 2026 – Present

- Interview Asian restaurant owners and managers to identify challenges with current AI systems, including authentic dish names, regional accents, multilingual communication, and technology adoption.
- Translate research findings into the design of an AI-powered restaurant management app that adapts to owners' languages and cultural needs rather than requiring restaurants to adapt to existing technology.

- Currently prototyping features including multilingual menu management, cross-platform ordering integration, automated translation, cost/profit analysis, culturally aware menu recommendations, and AI-assisted marketing tools.

EXTRACURRICULAR ACTIVITIES

***Team Captain*, Vex Robotics** May 2024 – Present

- Captain a nationally competitive robotics team, leading engineering design, build strategy, project planning, task delegation, scouting, and competition preparation.
- Serve as lead on-field strategist, analyzing scouting data, robot capabilities, and alliance compositions to develop match strategies and make real-time decisions under pressure.
- Lead engineering documentation and mentor elementary/middle school VEX teams in design, programming, notebook development, and competition strategy.
- Helped lead the team to 4 consecutive VEX World Championship qualifications, earning 25+ Tournament Championships, 10+ Excellence Awards, 15+ Robot Skills Championships, 2 Texas Amaze Awards, and 2 Signature Event Finalist finishes.

***Social Director,* Science Olympiad** Aug 2022 – Present

- Manage communications and community-building for an 85+ member team, including social media, team events, competition logistics, travel itineraries, and the after-school café.
- Selected as one of 15 competitors from 85+ members to represent Bridgeland at the Texas State Tournament for four consecutive years.
- Competed in 30+ Science Olympiad events, earning 20+ medals, including 2nd Place in Tower at the Texas State Tournament and 7th Place in Bungee Drop at the Golden Gate Invitational, a nationally competitive tournament in California.

***Leadership Team Officer,* Technology Student Association (TSA)** Aug 2024 – Present

- Support chapter operations through executive communications, meeting presentations, competition logistics, member activities, and mentorship of younger competitors.
- Earned 10+ regional and 8+ Texas State awards, including 1st Place in three Unique to Texas engineering events.
- Represent the chapter in engineering, data science, photography, presentation, and design competitions at regional and state levels.

***Builders Club Advisor,* Key Club** Aug 2024 – Present

- Lead biweekly leadership and community-service programming for 50+ middle school students across two partner schools.
- Organize service projects and mentor students in leadership, teamwork, communication, and civic engagement.
- Help facilitate biweekly Key Club programming and volunteer coordination for 150+ high school members.

***Calculator Applications Officer,* Math UIL** Aug 2025 – May 2026

- Mentor students through weekly Calculator Applications practices, competition strategies, and preparation for invitational, regional, and state contests.
- Built a centralized resource system containing practice tests, study guides, announcements, schedules, and competition materials.
- Coordinated competition logistics and helped establish Math UIL's social media presence.

Rice Elite Tech Camp Summer 2025

- Applied engineering and programming concepts through hands-on STEM design challenges, developing technical solutions to real-world problems in a collaborative team environment.
- Explored emerging technologies including AI, robotics, and coding while strengthening problem-solving, leadership, and engineering design skills through project-based learning.

Code 2 College Jun 2026 – Present

- Attended 9 week web programming course including basic and advanced Python concepts; attended professional development workshops to improve time management, professional skills, and technical portfolio development.
- Developed multiple Python applications, including an interactive Hangman game, an Escape Room adventure game, and an inventory management system using functions, loops, dictionaries, file handling, and object-oriented programming principles.
- Applied software engineering best practices by debugging code, implementing user-friendly features, organizing projects into reusable functions, and collaborating through iterative project development.

***Outreach, Driver, & Builder*, FIRST Tech Challenge (FTC)** Aug 2022 – May 2024

- Built and tested robot mechanisms, competed as a driver, and analyzed opponents to support match strategy and alliance selection.
- Led STEM outreach and digital media while mentoring an FLL team; secured $5,000+ in sponsorships from local businesses and community organizations.
- Helped the team earn Connect Awards, Winning Alliance Champion, and 3× Alliance Captain honors at regional and national competitions.

***Leadership Team Member,* Pioneer Academics Alumni & Social Media Committee** Aug 2025 – Present

- Create educational content for Pioneer Academics' official social media platforms, sharing research opportunities, application guidance, productivity strategies, and insights from my own research journey to support prospective and current scholars.
- Mentor current Pioneer researchers through alumni workshops by providing guidance on research design, academic writing, mentor communication, and navigating the independent research process.

- Contributed content and outreach initiatives that reached the broader Pioneer Academics community, helping students gain confidence in pursuing research and maximizing their experience within the program.

Freelance Photographer Aug 2021 – Present

- Pursued photography as a personal hobby since the age of 10, using a Panasonic Lumix digital camera to capture portraits, cityscapes, nature, and everyday moments; managing a photography portfolio on VSCO, showcasing original work.
- Accept freelance photoshoot bookings for portraits and special occasions, working with clients to create personalized, high-quality photographs.
- Operate an eBay storefront to sell original photography prints, gaining experience in online sales, customer service, order fulfillment, and small business management.

WORK EXPERIENCE

***E-Commercial Resale Business,* YUBAI** Summer 2023, 2024, 2025, 2026

- Provided bilingual (English/Mandarin) customer support by answering inquiries, resolving order issues, and assisting customers throughout the purchasing process.
- Designed personalized thank-you cards, photographed products, and supported customer outreach to enhance the brand's online presence and customer experience.
- Assisted with e-commerce operations by managing QR code systems, troubleshooting technical issues, organizing orders, and supporting daily business operations.

***Restaurant Operations Assistant,* Fang's Bistro** June 2026 – Present

- Created and managed the restaurant's Instagram account, producing promotional content to increase community engagement and attract new customers.
- Assisted front-of-house operations by greeting customers, answering phones, processing orders, and coordinating DoorDash and third-party delivery services.
- Redesigned restaurant menus with improved layouts and presentation to create a more professional customer experience.

COMMUNITY SERVICE

***Founder & Photographer,* Freedom Frame** April 2026 – Present

- Founded a global storytelling initiative dedicated to empowering women by documenting and sharing the stories of entrepreneurs, leaders, artists, and changemakers through photography and digital media.
- Photograph and interview inspiring women, creating compelling visual stories that celebrate resilience, leadership, and innovation while amplifying underrepresented voices.
- Lead global outreach efforts by recruiting photographers, writers, and ambassadors to spotlight impactful women within their own communities and expand the initiative's international reach.
- Direct the organization's branding, photography, social media, partnerships, and strategic growth while fostering a worldwide community committed to inspiring the next generation of female leaders.

***Houston Chapter Head,* Between the Lines** April 2025 – Present

- Lead the Houston chapter of a student-led initiative that amplifies the stories of Asian and Asian American athletes, challenges stereotypes, and promotes representation in sports.
- Organize community pickleball sessions and a local tournament, coordinating participants, event logistics, athlete photography, and 3D-printed prizes.
- Photograph and feature local athletes and contribute written and visual content to the organization's annual magazine, highlighting their achievements, identities, and athletic experiences.

HONORS & AWARDS

VEX Robotics

- **Signature Event Tournament Finalist**, **Ranking #2/75 (National),** Space City 2025
- **Signature Event Tournament Finalist, #1/90 teams (National),** Robot Rodeo 2026
- **Signature Event Innovate Award, Ranking #15/65 (National),** Space City 2025
- **2× Texas Region 3 Amaze Award**, qualifying for the **VEX Robotics World Championship** *(State)* 2025, 2026
- **25+ Local Tournament Championships, (District)** 2024, 2025, 2026
- **10+ Local Excellence Award, (District)** 2024, 2025, 2026
- **15+ Local Skills Championships, (District)** 2024, 2025, 2026

Technology Student Association (TSA)

- **1st place Mechanical Engineering - Working Drawings of a Single or Multiple Parts with a Pictorial View (Unique to Texas Event), (State)** 2024
- **1st place Mechanical Engineering - Pictorial View Only of 6-12 Mechanical Parts (Unique to Texas Event), (State)** 2024
- **1st Place Mechanical Engineering - Pictorial View Only of 2-5 Mechanical Parts (Unique to Texas Event), (State)** 2024
- **5th Place Photography - People - Portrait (Unique to Texas Event), (State)** 2026
- **1st Place Finish Model of a Residential Structure not listed (Unique to Texas Event), (State)** 2026

- **TSA Lonestar Degree** 2025
- **TSA Texan Degree** 2025
- **1st place Mechanical Engineering - Working Drawings of a Single or Multiple Parts with a Pictorial View (Unique to Texas Event), (Regional)** 2024
- **1st place Mechanical Engineering - Pictorial View Only of 6-12 Mechanical Parts (Unique to Texas Event), (Regional)** 2024
- **1st Place Mechanical Engineering - Pictorial View Only of 2-5 Mechanical Parts (Unique to Texas Event), (Regional)** 2024
- **1st Place Data Science and Analytics (National Qualifying Event), (Regional)** 2025
- **3rd Place Senior Solar Sprint (National Qualifying Event), (Regional)** 2025
- **1st Place Finish Model of a Residential Structure not listed (Unique to Texas Event), (Regional)** 2026
- **1st Place Prepared Presentation (National Qualifying Event), (Regional)** 2026
- **3rd Place Photography - People - Children (Unique to Texas Event), (Regional)** 2026
- **1st Place Photography - People - Portrait (Unique to Texas Event), (Regional)** 2026

Science Olympiad

- 2nd **Place, Tower** *(Texas State Tournament; 2nd of 50 teams)* 2025
- 7th **Place, Bungee Drop** *(National Invitational; 7th of 70 teams)* 2026
- 5x **Regional Placements** (Different Events) *(Regional)* 2024, 2025, 2026
- 8x **Regional Placements** (Different Events) *(District)* 2024, 2025, 2026
- 15x **Local Placements** (Different Events) *(District)* 2024, 2025, 2026

First Tech Challenge (FTC)

- **Winning Alliance Champion** *(Qualify for Regionals)* **(District)** 2024
- **Connect Award (District)** 2024
- **Finalist of Regional Championship** *(Qualify for State)* **(Regional)** 2024

USACO Bronze Division *(National)* 2025

College Board AP Scholar with Distinction 2025, 2026

SKILLS

Languages: English (Native), Mandarin Chinese (Native)
Programming: Python, Java, C++, OpenAI API Integration
Engineering: Autodesk Inventor, Fusion 360, AutoCAD, Prompt Engineering